PePNet: Pose-Enhanced Point Cloud Network for LiDAR-Based Human Action Recognition in Outdoor Long-Range Scenarios

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***Abstract*—With potential applications in robotics and autonomous vehicles, LiDAR-based human action recognition (HAR) in outdoor long-range scenarios is challenging due to the degradation of point cloud density with distance and the simultaneous motion of humans and sensors. To address these issues, we propose the Pose-Enhanced Point Cloud Network (PePNet), a distance-aware framework for long-range HAR. As the core component, the Pose-Enhanced Point Cloud Block (PeP Block) integrates three modules: a Dynamic Enhancement Module that mitigates point cloud sparsity at long distances by generating supplementary points from motion cues, a Pose Prompter Module that introduces pose priors, and an Adaptive Point Selection Module that suppresses irrelevant body-part movements. We further design a Spatiotemporal Tube Embedding (ST-Tube), combined with the Mamba state space model, to capture long-range dependencies and complex motion dynamics. In addition, we construct Momo, a large-scale LiDAR-based HAR dataset that focuses on long-range (2–30 m) outdoor scenarios where sparse point clouds and simultaneous human-sensor motion pose prominent challenges, complementing existing benchmarks by providing a dedicated evaluation platform for long-range outdoor HAR. Experimental results show that PePNet achieves consistent performance gains over existing methods on Momo. Moreover, the proposed PeP Block can serve as a plug-and-play module to enhance other point cloud action recognition frameworks in long-range outdoor settings. The code is available at https://github.com/Shark0-0/PePNet**

***Index Terms*—Action recognition, LiDAR data, point cloud.**

I. Introduction

HUMAN Action Recognition (HAR) is pivotal for applications in autonomous driving and intelligent surveillance systems [1], [2], [3], [4], [5], [6], [7]. LiDAR-based methodologies are gaining considerable attention due to their ability to capture precise 3D geometric information and maintain robustness against illumination variations [8], [9]. In outdoor long-range scenarios, LiDAR-based HAR enables critical downstream applications, such as early pedestrian intent prediction and behavior understanding for autonomous vehicles, where subjects are typically located tens of meters away from the sensor [10], as illustrated in Fig. 1. A critical factor frequently overlooked in LiDAR-based HAR is the impact of the distance between the sensor and the human subject. The quality and density of LiDAR point clouds deteriorate quadratically with increasing distance, posing a fundamental challenge to recognition accuracy, particularly in large-scale, dynamic environments where LiDAR technology is most valuable.

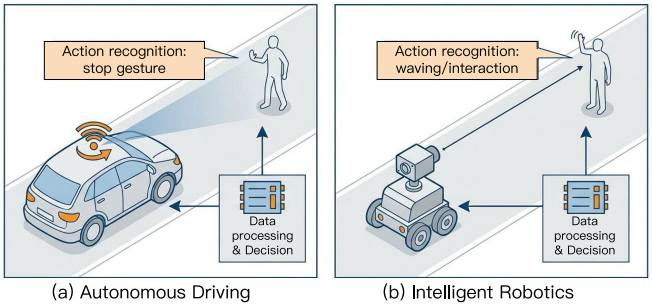


Fig. 1. Downstream applications of outdoor long-range LiDAR-based HAR: (a) autonomous driving and (b) intelligent robotics.

To directly address the critical role of distance, we introduce Momo: a novel LiDAR-based Motion-in-Motion dataset. Momo is explicitly designed to investigate and address the challenges posed by long-range perception. As shown in Fig. 2, the most obvious challenge posed by Momo lies in its extreme point cloud density inconsistency caused by varying subject distances. As the distance increases, point clouds become sparser and lose critical structural details; conversely, nearby subjects produce dense representations that demand careful processing. This variation in density introduces instability in feature extraction and action modeling. Moreover, the motion-in-motion setup introduces complex temporal shifts and dynamic viewpoints, making spatiotemporal modeling even more difficult.

To address the aforementioned challenges, we propose the Pose-Enhanced Point Cloud Block (PeP Block), a dedicated module targeting the core difficulties of the Motion-in-Motion setting. It comprises three complemen-

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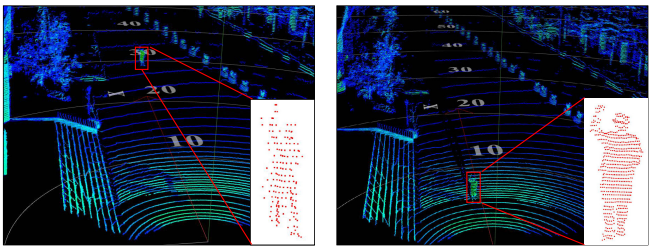
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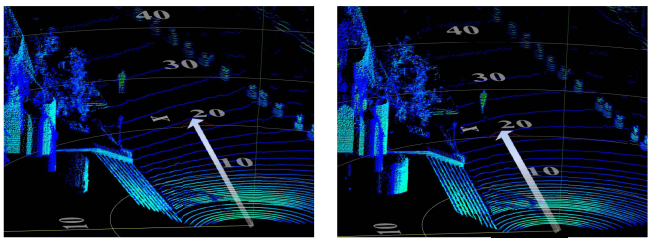
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(a) Human motion



(b) Sensor motion

Fig. 2. Illustration of the core challenges in the Momo dataset. (a) Human motion: as the subject moves away from the sensor, point cloud density degrades significantly. The red bounding boxes highlight the same individual at two different distances, with insets showing the corresponding sparse (far) and dense (near) point cloud representations. (b) Sensor motion: the white arrows indicate the moving direction of the LiDAR sensor, which causes continuous viewpoint shifts across frames. The coupling of human and sensor motion constitutes the "Motion-in-Motion" (MiM) challenge, introducing dynamic viewpoints and temporal inconsistency.

tary components: (i) a Dynamic Enhancement module that recovers missing geometric details under distance-induced sparsity, (ii) a Pose Prompter that injects skeleton-level geometric priors to preserve shape consistency across varying observation ranges, and (iii) an Adaptive Point Selection module that retains the most action-discriminative enhancement points while suppressing redundant contributions from less informative body regions. We further introduce Spatiotemporal Tube Embedding (ST-Tube), which partitions sequences into structured motion tubes for stable and localized temporal modeling, and adopt a Mamba-based architecture to capture long-range temporal dependencies efficiently.

Our framework integrates three core components: (1) a pose-guided enhancement mechanism that dynamically augments sparse regions, (2) Spatiotemporal Tube Embedding (ST-Tube) for effective temporal modeling, and (3) Mamba Block that captures complex spatiotemporal dynamics while maintaining robustness to viewpoint and distance variations inherent in challenging datasets like Momo. This comprehensive approach ensures consistent action recognition performance regardless of subject proximity to the sensing device. In summary, our main contributions are threefold:

- We present the Momo dataset, a large-scale LiDAR-based HAR dataset that emphasizes long-range interactions (2 m to 30 m) and incorporates a "motion-in-motion" setting, where both subjects and sensors are in motion. The dataset is specifically designed to facilitate the study of challenges related to distance variation in complex outdoor environments.
- We propose PePNet, a distance-aware framework that addresses the issue of inconsistent point cloud density caused by large distance variations. To capture local motion patterns, we introduce ST-Tube, which encodes spatiotemporal structures into tube-like segments. In addition, we utilize Mamba blocks to model long-range dependencies in point cloud sequences while maintaining computational efficiency.
- We evaluate our method on the Momo dataset and observe consistent performance improvements compared to prior approaches. The results suggest that PePNet can serve as a modular component that improves robustness to distance-induced sparsity and may benefit existing architectures, particularly in long-range scenarios where point cloud data is sparse and noisy.

II. Related Work

A. Datasets for LiDAR-Based Human Action Recognition

The advancement of LiDAR-based Human Action Recognition (HAR) critically depends on the availability of suitable datasets. While early benchmarks such as NTU RGB + D [12], [18], Behave [14], and DFAUST [19] offer valuable depth or pose annotations, they are typically limited to controlled indoor settings and lack native LiDAR data. Recent datasets like LiDARCap [20], HuCenLife [16], and LIP [21] introduce human-centric LiDAR point clouds, but are not explicitly designed for diverse action recognition in dynamic, long-range scenarios. As summarized in Table I, existing benchmarks often involve fixed sensors, short interaction distances, or focus on tasks other than outdoor HAR. A clear gap remains in datasets tailored for long-range (e.g., up to 30m), dynamic HAR where both humans and sensors may be in motion. Our proposed Momo dataset directly addresses this gap by providing realistic, action-diverse, and sensor-dynamic LiDAR data for outdoor HAR.

B. Methods for Point Cloud Modeling

Effective modeling of dynamic point cloud sequences requires capturing both intra-frame spatial geometry and inter-frame temporal dynamics. Spatial modeling typically leverages hierarchical feature learning over local neighborhoods, as in PointNet++ [22], or dynamic graph structures like DGCNN [23]. For spatio-temporal modeling, MeteorNet [24] extends PointNet++ to jointly encode space-time features, while PSTNet [9] separates spatial and temporal streams to enhance motion learning. Multi-stream models such as 3DV [25] and GMTNet [26] integrate voxelized motion cues and temporal fusion modules. Transformer-based methods like P4Transformer [27] and STPT [28] employ attention mechanisms for long-term temporal reasoning.

State space models have emerged for efficient long-range sequence modeling. Mamba [29], a selective state space model, offers linear scalability and implicit recurrence, making it effective for long-context tasks. Vision Mamba [30] demonstrates that sequence ordering is a critical design choice

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TABLE I
COMPARISON OF OUR PROPOSED MOMO DATASET WITH RELATED DATASETS

Dataset	Year	Subjects	Scenes	Data Types	Samples	Distance	Outdoor	Motion	
								Human	Sensor
MSRAction3D [11]	2010	10	1	Depth camera	567 videos	<5m	✗	✗	✗
NTU RGB+D 120 [12]	2019	106	32	Depth camera	114,480 videos	2-5m	✗	✗	✗
UAV Human [13]	2021	119	1	Depth camera	67,428 videos	-	✗	✗	✗
BEHAVE [14]	2022	8	5	Depth camera	∼15,000 frames	-	✗	✓	✗
IKEA ASM [15]	2019	48	5	Depth camera	371 samples	-	✗	✓	✗
HuCenLife [16]	2023	-	Diverse	LiDAR	27,142 videos	-	✓	✓	✗
HmPEAR [17]	2024	25	10	LiDAR	300,000 frames	3-15m	✓	✓	✗
Momo (Ours)	2025	80	7	LiDAR	45,000 videos	**2-30m**	✓	✓	✓

for SSM-based visual models, introducing bidirectional scanning over spatially reordered image patches to capture global context. PointMamba [31] extends this insight to 3D point clouds by incorporating geometry-aware spatial reordering to inject structural bias into the SSM pipeline. Mamba4D [32] further advances this line of work to the spatio-temporal domain, disentangling spatial and temporal state spaces via gating mechanisms for 4D point cloud video understanding. Recent studies further cover LiDAR HAR spectral descriptors [33], multimodal point cloud–skeleton–IMU–text pre-training [34], lightweight 4D video backbones [35], and dynamic point cloud understanding [36]. Despite these advancements, existing methods adopt scanning strategies designed for general 3D geometry without specifically exploiting the structured longitudinal organization of the human body, nor do they address the severe distance-dependent density variation inherent in outdoor long-range LiDAR HAR. In contrast, our method introduces a pose-informed adaptive density normalization mechanism and a Z-axis prioritized longitudinal scanning strategy that aligns SSM state transitions with the natural head-to-toe anatomical structure, enabling robust action recognition under large depth variations in outdoor scenarios.

III. MOMO DATASET

To overcome the limitations of existing benchmarks, primarily focused on constrained or static settings, we construct Momo, a large-scale LiDAR-based dataset specifically designed for human action recognition in complex Motion-in-Motion scenarios typical of real-world environments. Momo embodies the principles of "realistic scenes, diverse interactions, and large-scale data," providing a challenging benchmark for developing and evaluating robust HAR algorithms. This section details the dataset preparation, annotation process, and statistical characteristics.

A. Data Acquisition

The dataset preparation phase encompassed several critical stages to ensure realism and comprehensiveness. Central to this was the sensor selection process. We required a sensor capable of capturing human actions across both short (2-5m) and long distances (20-30m) reliably in complex, uncontrolled environments. Comparing various 3D sensor modalities, we found LiDAR to be superior to RGB-D cameras (limited range), structured light sensors (poor performance under strong ambient illumination), and millimeter-wave radar (sparse point clouds unsuitable for detailed pose). LiDAR's advantages in range, accuracy, stability, and environmental adaptability made it the ideal choice. Specifically, we selected the Hesai AT128P, a 128-line hybrid solid-state LiDAR, suitable for human-centric data acquisition in diverse environments. Following sensor selection, we designed a comprehensive set of 30 action classes. These actions represent common human behaviors in real-world scenarios, which are critical for autonomous systems and robotics. As summarized in Table II, we organize the 30 action classes into three structured categories: *Individual Behaviors* (e.g., walking, stretching), *Social Interactions* (e.g., waving, nodding), and *Path Adjustments* (e.g., turning, pointing). Notably, most actions are performed concurrently with walking, reflecting realistic composite movement patterns. To ensure environmental diversity, data collection occurred across seven distinct scenes (Fig. 3), including locations like walkways, underground parking garages, corridors, and roadsides, captured under both daytime and nighttime conditions. A key aspect of our design was the data collection protocol involving specific routes and motion patterns. Participants followed a predefined polyline path starting approximately 30m from the sensor, incorporating turns and changes in direction to induce continuous variation in distance and viewing angle while performing actions. Critically, the sensor platform itself

TABLE II
ACTION CLASSIFICATION TAXONOMY OF THE MOMO DATASET, CATEGORIZED INTO THREE DISTINCT BEHAVIOR TYPES: INDIVIDUAL, SOCIAL, AND PATH-ADJUSTMENT

Individual Behaviors	Social Interactions	Path Adjustments
1. Walk	11. Walk + Wave	21. Walk + Clockwise turn
2. Walk + Raise hands	12. Walk + Handshake	22. Walk + Counter-clockwise turn
3. Walk + Rub hands	13. Walk + Reach out	23. Walk + Move left
4. Walk + Scratch head	14. Walk + Nod	24. Walk + Move right
5. Walk + Touch chin	15. Walk + Shake head	25. Walk + Move up
6. Walk + Fold arms	16. Walk + Clap	26. Walk + Move down
7. Walk + Hands on hips	17. Walk + Thumbs up	27. Walk + Look left
8. Walk + Stretch	18. Walk + Thumbs down	28. Walk + Look right
9. Walk + Shrug	19. Walk + Phone call	29. Walk + Beckon
10. Walk + Drink water	20. Walk + Texting	30. Walk + Repel

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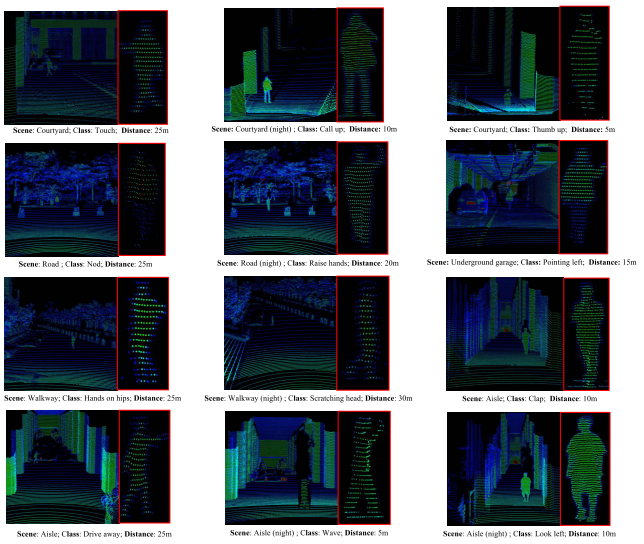


Fig. 3. Illustrative LiDAR point cloud samples for human action recognition. Each sub-figure presents a full scene (left) alongside a magnified view of the human subject (right, red box). Annotations detail the recording scene (e.g., Courtyard, Road, Aisle, under day/night conditions), the action class performed (e.g., Touch, Call up, Nod, Wave), and the subject's distance (5m to 30m). These examples highlight the variability in LiDAR data across diverse actions and distances, emphasizing the critical role of distance.

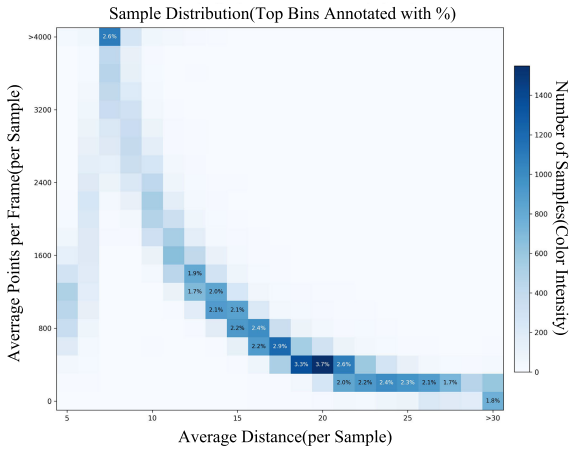


Fig. 4. Distribution of Momo dataset samples based on their average distance (distance from sensor, m) and average number of points per frame. Cell color intensity corresponds to the number of samples in each 2D bin (darker indicates more samples). Annotations indicate the percentage of the total analyzed samples (Distance ≥ 5m) falling into the 20 most populated bins, illustrating the pronounced trend of decreasing point cloud density with increasing distance.

was mobile, simulating real-world mobile sensing platforms. This simultaneous movement of both subject and sensor establishes the core "Motion-in-Motion" characteristic of Momo, introducing dynamic viewpoints and relative coordinate shifts, significantly increasing the challenge and realism compared to static sensor setups.

B. Dataset Annotation

Subsequent to data collection, a rigorous annotation pipeline was employed to process the raw LiDAR sequences into labeled action samples suitable for HAR tasks. The initial data capture involved 80 subjects performing multiple rounds of walking along the designated routes, executing 15-20 specific actions per round based on verbal cues from an experimenter who simultaneously recorded precise start and end timestamps for each action. These timestamps formed the basis for the first processing step: video segmentation. The continuous LiDAR recordings were temporally segmented using the recorded timestamps, creating individual clips where each clip corresponds to a single action instance, thus ensuring clear temporal boundaries. For each segmented action sample, we then applied 3D object detection to localize the human subject within every frame. We utilized the BEVFusion algorithm [37], generating accurate 3D bounding boxes around the subject even in cluttered and dynamic scenes. Following detection, the human point cloud was extracted. To account for movement during the action, we determined the maximum spatial extent covered by the subject's bounding boxes across all frames within a sample and extracted all points falling within this aggregated volume. This ensured that the complete motion was captured. Finally, the extracted point clouds underwent a two-stage refinement process. Ground points were removed to reduce background clutter and computational load, followed by a clustering-based denoising step to filter outliers and enhance the integrity of the human body representation. This systematic pipeline resulted in 45,000 high-quality, annotated action samples, each containing a sequence of refined point clouds corresponding to a specific action label.

C. Dataset Statistics

The Momo dataset presents key features that make it a valuable yet challenging benchmark for Human Action Recognition (HAR). It contains 45,000 samples from 80 subjects with diverse demographics such as gender, physique, and clothing, captured across 7 scenes under varied lighting. It covers 30 action classes with approximately 1,600 samples per class. A distinctive aspect is its long-range capture capability of up to 30 meters, which results in significant variation in point cloud density. As shown in Fig. 4, point density decreases with increasing distance. Subjects within 5 to 10 meters may have over 4,000 points per frame, while samples beyond 20 meters often contain fewer than 200 points per frame. This distance-related sparsity presents a core challenge for recognition.

As illustrated in Fig. 5, the contrast between Momo and indoor benchmarks is striking. In MSRAction3D and NTU RGB + D 120, subjects are represented by dense, structurally complete point clouds where body contours and limb boundaries are clearly distinguishable. In Momo, however, the LiDAR scanning pattern manifests as discrete horizontal ring layers, and at long ranges the human body is reduced to a sparse set of isolated points — often fewer than 200 per frame — where even coarse body shape becomes difficult to resolve. Beyond density, the nature of the actions themselves differs fundamentally: Momo samples require models to jointly parse simultaneous leg locomotion and subtle upper-body gestures, such as waving while walking, whereas indoor datasets typically isolate a single action performed

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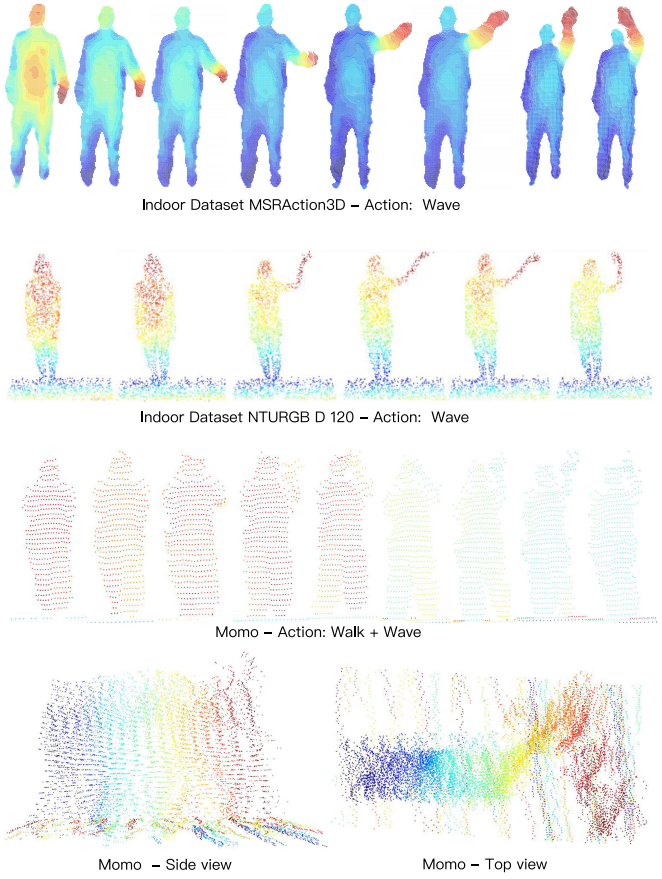


Fig. 5. Comparison of point cloud sequences between Momo and indoor datasets (MSRAction3D, NTU RGB + D 120).

by a stationary subject. Moreover, point colors in the Momo sequence encode depth along the LiDAR beam direction, and a notable color discontinuity is observable at the sixth frame, reflecting an abrupt change in the relative sensor–subject geometry caused by the simultaneous motion of both the LiDAR platform and the human subject. This phenomenon is further corroborated by the side and top views in Fig. 5, which reveal the non-linear spatial trajectories and continuous viewpoint drifts characteristic of the Motion-in-Motion setting. Such challenges are entirely absent in indoor datasets with fixed sensors, making Momo a considerably more demanding benchmark than its indoor counterparts.

Compared to existing datasets in Table I, Momo fills an important gap. Previous datasets often focus on controlled indoor environments (such as NTU RGB + D), object detection in autonomous driving scenarios (such as KITTI), or high-precision motion capture (such as LiDARCap). In contrast, Momo integrates large scale, diverse subjects and scenes, composite actions, long interaction ranges, and the Mobile-in-Mobile (MiM) setting where both the subject and the sensor are moving. These characteristics require models to handle dynamic scenes with shifting reference frames, recognize actions reliably under extreme sparsity, and manage wide density variations across samples. Momo provides a comprehensive benchmark that encourages the development of robust and generalizable HAR methods for real-world dynamic environments.

D. Point Cloud Sparsity and Distance-Dependent Density

A big challenge in LiDAR-based human action recognition arises from the inherent sparsity of point clouds, which increases significantly with distance. This phenomenon is primarily due to the geometric nature of LiDAR scanning and the inverse-square law governing laser intensity distribution.

1) Geometric Spreading Effect: LiDAR sensors emit laser beams at a fixed angular resolution (e.g., 0.1° per beam), forming a fan-like or cone-like coverage pattern as the sensor rotates. As the distance r between the sensor and the target increases, the surface area A covered by each beam grows quadratically:

$$A(r) = 2\pi r^2 \Delta\theta, \tag{1}$$

where $\Delta\theta$ is the angular resolution of the beam. Assuming a constant number of points per beam, the point density D (points per unit area) decreases with the square of the distance:

$$D(r) \propto \frac{1}{r^2}. \tag{2}$$

This leads to a rapid drop in point cloud density as the subject moves farther away from the sensor — a key factor contributing to the sparse representation of distant humans in our dataset.

2) Signal Attenuation and Noise: In addition to geometric spreading, signal attenuation due to atmospheric absorption and scattering further reduces the effective return strength of the laser pulses. The received power P_r at distance r can be modeled as:

$$P_r(r) = P_0 \cdot e^{-\alpha r} \cdot \left(\frac{d}{2r}\right)^2. \tag{3}$$

IV. Method

Before detailing our method, we briefly review the foundational concepts utilized: 3D Human Pose Estimation from point clouds and State Space Models (Mamba).

a) *3d human pose estimation from point clouds:* Human Pose Estimation aims to identify the spatial locations of key body joints (e.g., shoulders, elbows, knees) [38], [39]. While traditionally performed on 2D images [40], 3D pose estimation provides crucial spatial information, overcoming the depth ambiguity of 2D methods. When applied to point clouds $P = \{p_1, \ldots, p_N\} \subset \mathbb{R}^3$, the goal is to infer the 3D coordinates of a set of K joints $J = \{j_1, \ldots, j_K\} \subset \mathbb{R}^3$ [41], [42], [43], [44], [45], [46], [47]. Deep learning methods like PointNet [48] and PointNet++ [22] enable direct processing of unstructured point data. Typically, point cloud-based pose estimation involves preprocessing (denoising, segmentation), feature extraction using deep networks, and joint regression. Formally, a model f parameterized by θ maps the input point cloud P to the joint set J:

$$J = f(P; \theta). \tag{4}$$

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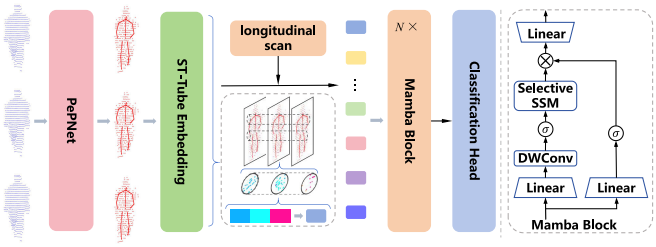


Fig. 6. Overview of our framework for LiDAR-based human action recognition. The pipeline consists of four components: 1) Human Pose Enhancement module that normalizes point cloud density across varying distances, 2) ST-Tube Embedding that constructs spatiotemporal sequences from enhanced point representations, 3) Mamba Block for effective modeling of complex motion patterns, and 4) Classification Head for final action recognition.

Training often minimizes a loss function like Mean Squared Error (MSE) between predicted joints j_k and ground truth joints $\hat{j}_k$:

$$L = \frac{1}{K}\sum_{k=1}^{K} \|j_k - \hat{j}_k\|^2. \tag{5}$$

In our work, we utilize pre-computed 3D poses as prior knowledge.

b) *State space models (Mamba):* Mamba [29], [49] is a recent class of sequence models based on Structured State Space Models (SSMs) that offer linear complexity scaling with sequence length, contrasting with the quadratic complexity of Transformers. A continuous SSM maps an input function $x(t)$ to an output $y(t)$ via a latent state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}\mathbf{h}(t) + \mathbf{B}x(t), \tag{6}$$

$$y(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}x(t), \tag{7}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are state matrices. Mamba employs selective SSMs (S4) where these matrices become input-dependent, allowing the model to focus on relevant information selectively. It utilizes efficient parallel scan algorithms for hardware-aware computation. Mamba has shown strong performance on long sequence tasks and has been adapted for vision (Vision Mamba [30]) by treating image patches as sequences. Recent works have started exploring Mamba for 3D point clouds [31], [32], [50], [51], requiring careful integration with spatial feature extractors. Given a point cloud sequence $\mathcal{V} = \{\mathbf{P}_t, \mathbf{F}_t\}_{t=1}^{L}$, a typical Point Cloud Mamba pipeline first extracts per-frame spatial features $\mathbf{f}_t = f(\mathbf{P}_t, \mathbf{F}_t)$ and then models the temporal dependencies using Mamba:

$$\text{feature}_{\text{Mamba}} = \text{Mamba}([\mathbf{f}_1, \ldots, \mathbf{f}_L]). \tag{8}$$

A. PeP Block

A critical challenge in recognizing actions over long ranges (as in Momo) is the inconsistency in point cloud density due to varying subject distances. Near subjects yield dense clouds, while far subjects yield sparse ones, hindering consistent feature extraction. To mitigate this, we propose the Pose-Enhanced Point (PeP) Block, illustrated in Fig. 7. It leverages an estimated 3D human pose as prior knowledge to dynamically augment the point cloud, aiming for improved density consistency and completeness, especially in sparse regions corresponding to body parts. The module takes the original point cloud and its corresponding 3D pose estimate as input and outputs an enhanced point cloud representation. It consists of a dynamic enhancement algorithm and an adaptive point selection mechanism guided by a pose prompter.

1) Dynamic Enhancement Module: Instead of static upsampling, this algorithm generates enhancement points dynamically along the skeletal edges defined by the input pose $P \in \mathbb{R}^{T\times J\times 3}$ (where T is sequence length, $J = 15$ is the number of joints). Given the predefined skeletal connectivity $E = \{(s_i, e_i)\}$, defining N_E edges, we extract the start $\mathbf{p}_{s_i}$ and end $\mathbf{p}_{e_i}$ points for each edge i. We introduce learnable parameters $\mathbf{t}_{\text{params}} \in \mathbb{R}^{N_E\times N_P\times 1}$ representing relative positions along each edge (where N_P is the number of points per edge). These are mapped to $(0, 1)$ using $\mathbf{t} = \sigma(\mathbf{t}_{\text{params}})$. To adapt to varying bone lengths during motion, we compute normalized lengths

$$\mathbf{L}_i = \|\mathbf{p}_{e_i} - \mathbf{p}_{s_i}\|_2 / \max_j(\|\mathbf{p}_{e_j} - \mathbf{p}_{s_j}\|_2), \tag{9}$$

and scale the parameters:

$$\mathbf{t}_{\text{scaled}} = \mathbf{t} \cdot \mathbf{L}_i / (\max_j(\mathbf{t} \cdot \mathbf{L}_j) + \epsilon). \tag{10}$$

The N_P new points $\mathbf{p}_{\text{new}}$ for edge i are then generated via linear interpolation:

$$\mathbf{p}_{\text{new}} = (1 - \mathbf{t}_{\text{scaled}}) \cdot \mathbf{p}_{s_i} + \mathbf{t}_{\text{scaled}} \cdot \mathbf{p}_{e_i}. \tag{11}$$

This allows the enhancement process to adapt dynamically to the current pose and bone scaling.

2) Pose Prompter: Recognizing that different body parts contribute differently to various actions, we introduce a Pose Prompter to generate importance weights for each skeletal edge based on the pose itself. For each edge i, the start and end point coordinates are concatenated and fed through an MLP to produce an edge weight

$$\mathbf{w}_{\text{edge}} = \sigma(\text{MLP}_{\text{edge}}(\text{concat}(\mathbf{p}_{s_i}, \mathbf{p}_{e_i}))) \in \mathbb{R}^{T\times N_E\times 1}. \tag{12}$$

This weight reflects the learned importance of the edge for the action recognition task. The generated points $\mathbf{p}_{\text{new}}$ are then augmented with this weight:

$$\mathbf{f}_{\text{point}} = \text{concat}(\mathbf{p}_{\text{new}}, \mathbf{w}_{\text{edge}}) \in \mathbb{R}^{T\times N_E\times N_P\times 4}. \tag{13}$$

This feature vector combines geometric position (first 3 dimensions) with semantic importance (4th dimension).

3) Adaptive Point Selection Module: Generating many enhancement points can introduce redundancy. We employ an Adaptive Point Selection mechanism, termed Pose Prompted Sampling (PPS), to select a representative subset ($M = 512$ points in our case) from the generated points $\mathbf{f}_{\text{point}}$ (reshaped to $N\times C$, where $N = T\times N_E\times N_P$ and $C = 4$). This mechanism, depicted in Fig. 7, consists of feature encoding, fusion, and differentiable sampling. First, local point features are encoded using an MLP (MLP_1, implemented with 1D convolutions and

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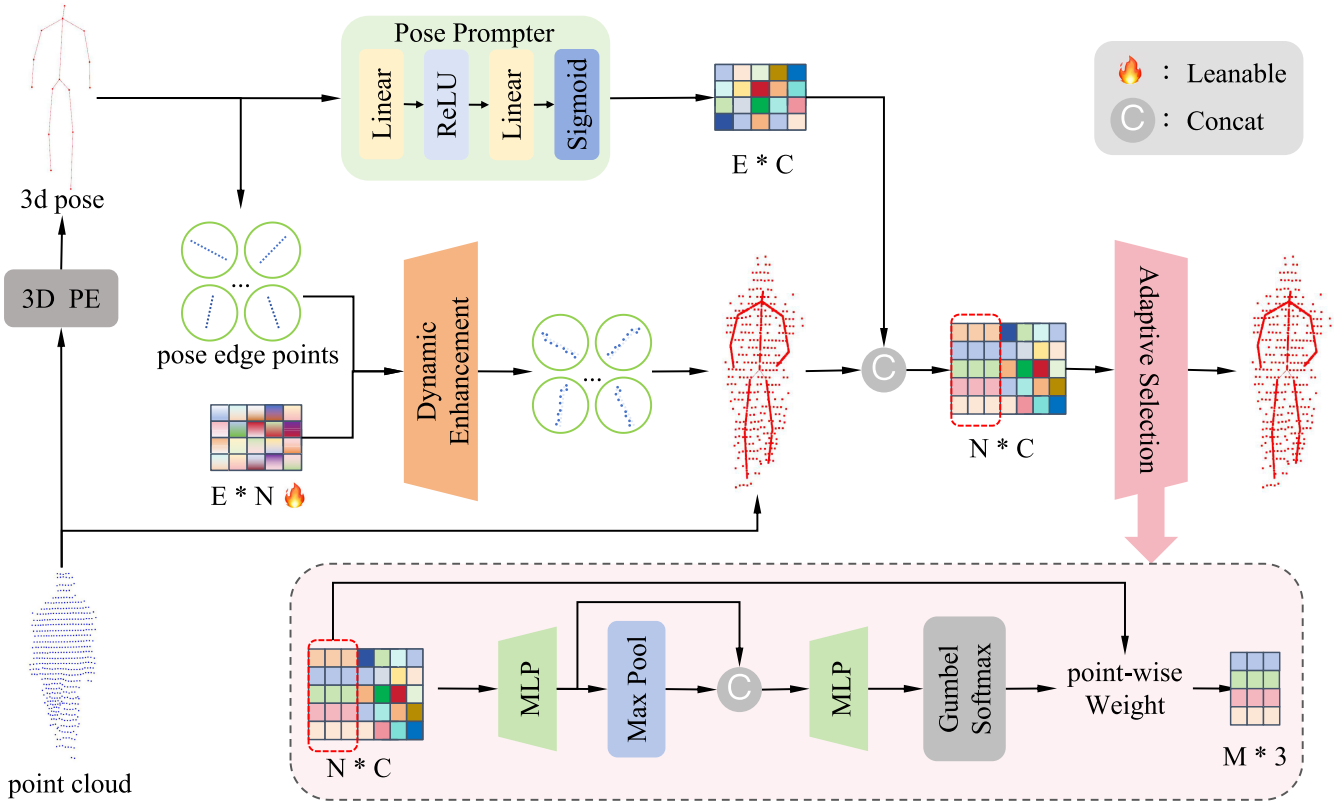


Fig. 7. Overview of the Pose Enhancement Point module (PeP block). PeP block takes as input a raw 3D point cloud and an estimated human pose. From the pose, dynamic enhancement points are generated along skeletal edges via learnable interpolation, enabling adaptive densification of body regions. A Pose Prompter assigns importance weights to skeletal edges based on their relevance to the action, embedding this semantic cue into the augmented points. Finally, the Adaptive Selection mechanism adaptively selects the most informative enhancement points using a fused feature representation and Gumbel-Softmax-based differentiable sampling.

BatchNorm). Global context is obtained via max pooling and concatenated with the local features:

$$\mathbf{y}' = \text{Concat}[\text{MLP}_1(\mathbf{X}), \text{MaxPool}(\text{MLP}_1(\mathbf{X}))]. \tag{14}$$

This fusion allows selection based on both local detail and global context. The fused features are processed by another MLP (MLP_2 with BatchNorm and ReLU) to extract higher-level semantics. Sampling probabilities are then computed using the Gumbel-Softmax function:

$$\mathbf{prob} = \text{GumbelSoftmax}(\text{MLP}_2(\mathbf{y}'), \tau), \tag{15}$$

where τ is the temperature parameter, annealed during training. Gumbel-Softmax provides a differentiable approximation to discrete sampling, enabling end-to-end training. Finally, the selected points $\mathbf{p}_{\text{sample}} \in \mathbb{R}^{M\times 3}$ are obtained by a weighted reconstruction using the probabilities:

$$\mathbf{p}_{\text{sample}} = \mathbf{prob} \cdot \mathbf{X}[:, :3, :]^T. \tag{16}$$

This differentiable mapping selects the M most representative points based on learned probabilities. The final enhanced point cloud is formed by concatenating the original points $\mathbf{p}_{\text{original}}$ with the selected enhanced points $\mathbf{p}_{\text{sample}}$:

$$\mathbf{p}_{\text{enhanced}} = \text{concat}(\mathbf{p}_{\text{original}}, \mathbf{p}_{\text{sample}}). \tag{17}$$

This preserves original information while selectively densifying key areas identified via pose priors.

B. ST-Tube Embedding

The ST-Tube Embedding module is responsible for converting enhanced point cloud sequences into structured spatiotemporal representations. This process consists of two main steps: spatiotemporal anchor point selection and spatiotemporal tube construction.

1) Spatiotemporal Anchor Point Selection: To capture multi-granularity spatiotemporal information, we first select anchor frames based on temporal stride s_t along the temporal dimension. Given a point cloud video sequence $\{[P_t, F_t], P_t \in \mathbb{R}^{N\times 3}, F_t \in \mathbb{R}^{N\times C}\}_{t=1}^{T}$, where P_t and F_t represent the point coordinates and features of the t-th frame respectively, the entire video sequence is divided into $\frac{T}{s_t}$ equal-length clips, with each clip constructed around an anchor frame using temporal kernel size k_t. In the spatial dimension, we employ Farthest Point Sampling (FPS) strategy to downsample each anchor frame with spatial downsampling rate s_s, obtaining $n = \frac{N}{s_s}$ spatial anchor points.

2) Spatiotemporal Tube Construction: For each spatial anchor point, we search for K-nearest neighbors (KNN) within its temporal clip, constructing spatiotemporal neighborhoods

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centered around the anchor point with spatial radius k_s. These cross-frame neighboring points form tube-like spatiotemporal intervals called point tubes. Each point tube contains local spatiotemporal geometric structure information, providing structured input for subsequent spatiotemporal modeling.

C. Mamba Block

The Mamba Block is the core component of our framework, designed based on State Space Models (SSM) for efficiently modeling long-sequence spatiotemporal dependencies. Compared to traditional Transformers, Mamba Block has the advantage of linear complexity, making it particularly suitable for processing long-sequence point cloud video data.

Inspired by the longitudinal structure of the human body, we adopt a human body longitudinal structure scanning approach in the Mamba block. Specifically, we design spatiotemporal scanning strategies that prioritize the vertical (Z-axis) direction to align with the natural longitudinal structure of human poses. The scanning process consists of two main strategies:

1) Human Body Longitudinal Scanning: We first spatially order all input point frames according to their vertical coordinates (Z-axis), which corresponds to the human body's longitudinal structure from head to toe. Given point coordinates (x, y, z), we sort points primarily by their Z-coordinate values:

$$\text{Sort}(\{P_i\}_{i=1}^{N}) = \text{argsort}(\{z_i\}_{i=1}^{N}). \tag{18}$$

2) Bidirectional Temporal Scanning: After spatial ordering along the longitudinal direction, we employ two complementary scanning strategies: 1) Temporally Sequential Scanning - points are scanned following the temporal order while maintaining the longitudinal spatial ordering; 2) Cross-temporal Scanning - points are scanned across different time frames to establish long-range temporal dependencies.

The mathematical formulation of Mamba block is:

$$\mathbf{h}_k^{(z)} = \mathbf{A}\mathbf{h}_{k-1}^{(z)} + \mathbf{B}\mathbf{x}_k^{(z)}, \tag{19}$$

$$\mathbf{y}_k^{(z)} = \mathbf{C}\mathbf{h}_k^{(z)}, \tag{20}$$

where $\mathbf{x}_k^{(z)}$ represents the Z-axis ordered input sequence, and the superscript (z) indicates the longitudinal scanning direction.

Each Mamba Block contains components including layer normalization (LN), SSM, depth-wise convolution, and residual connections:

$$F_l' = \text{DW}(\text{MLP}(\text{LN}(F_{l-1}))), \tag{21}$$

$$F_l = \text{MLP}(\text{LN}(\text{SSM}(\sigma(F_l'))) \times \sigma(\text{LN}(F_{l-1}))) + F_{l-1}. \tag{22}$$

This longitudinal scanning design enables the model to effectively capture the natural flow of human motion patterns while maintaining linear complexity, providing powerful feature representation capabilities for human action recognition. The Z-axis prioritized scanning particularly benefits human pose understanding by following the natural top-to-bottom structure of human body movements.

V. Experiments

This section systematically evaluates the effectiveness of the proposed PePNet and the Mamba-based spatio-temporal modeling algorithm. We first detail the experimental setup and evaluation protocols. Subsequently, a series of ablation studies are presented to analyze the contribution of each component. Finally, we provide a comprehensive comparison of our method against existing state-of-the-art (SOTA) point cloud action recognition techniques to demonstrate its advantages.

A. Implementation Details

1) Dataset: All experiments are conducted on two datasets: the proposed **Momo** dataset and the **Hmpear** dataset.

a) Momo dataset: Momo comprises point cloud sequences of human actions captured in diverse and complex environments, including challenging conditions like varying illumination and non-uniform point cloud density. It features 30 action classes, such as walking, waving, and turning, performed by 80 subjects across 7 scenes with distances ranging from 2 to 30 meters. The dataset is specifically designed to address the "motion-in-motion" scenario where both subjects and LiDAR sensors undergo compound movements, making it particularly challenging for real-world applications.

b) Hmpear dataset: The Hmpear dataset serves as an additional benchmark for evaluating our method's generalization capability. It contains human point cloud sequences captured in controlled environments with 43 action classes performed by 25 subjects. Unlike Momo, Hmpear features relatively stable sensor-subject distances (typically 3-15 meters) and controlled lighting conditions, providing a complementary evaluation setting that helps assess method robustness across different environmental complexities.

For both datasets, each point cloud frame within a sequence is uniformly downsampled or padded to contain 1024 points, represented by their 3D coordinates (x, y, z). The training/validation/testing splits follow the standard protocols established for each dataset.

2) Experimental Settings: The network is trained using the SGD optimizer with an initial learning rate of 0.001, which is decayed using a cosine annealing schedule. A weight decay of 10^{-4} is applied for regularization. All models are trained for 100 epochs on a single NVIDIA RTX 3090 GPU with a batch size of 32. Based on preliminary experiments, for the Human Pose Prior Enhancement module, we sample 512 points from the original point cloud and 512 points from the enhanced point cloud, finding this 1:1 ratio optimal for balancing original geometric detail and pose-guided structural information. Unless otherwise specified, the SSM-based sequence modeling uses Z-axis sorting combined with our proposed cross-temporal scan strategy.

B. Comparison With State-of-the-Art Methods

1) Quantitative Results: Table III presents a comprehensive comparison on both the Momo and Hmpear datasets. Our PePNet achieves 54.63% on Momo, outperforming all baselines. The strongest competitor, PointNet4D [35] (Mamba-Transformer hybrid), reaches 51.24%; PePNet surpasses it

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TABLE III

PERFORMANCE COMPARISON WITH SOTA METHODS ON THE MOMO AND HMPEAR DATASETS, INCLUDING DISTANCE-BASED ACCURACY BREAKDOWN FOR THE PROPOSED METHOD (OURS)

Method	Source	Momo Acc (%)	Hmpear Acc (%)	Accuracy by Target Distance (m)						
				<5	5–10	10–15	15–20	20–25	25–30	>30
Pose-based methods										
CTR-GCN [52]	ICCV'21	42.49	-	38.47	44.32	46.18	43.75	37.92	31.64	15.28
A^2MC [53]	TIP'26	34.94	-	32.89	39.31	40.52	37.53	27.65	21.74	8.97
Point cloud-based methods										
PointNet++ [22]	NIPS'17	36.86	41.73	33.52	40.15	42.37	39.64	34.28	28.93	13.86
PSTNet [9]	ICLR'21	44.36	50.14	40.23	46.58	48.72	46.14	40.37	33.95	16.75
PSTNet++ [54]	TPAMI'22	45.58	51.47	41.34	47.92	50.15	47.58	41.73	35.28	17.92
PST-Transformer [55]	TPAMI'23	45.64	61.04	39.29	48.69	49.07	45.75	40.45	34.26	17.53
3DInAction [56]	CVPR'24	46.23	-	40.56	49.88	50.14	47.35	42.73	35.91	13.26
Mamba4D [32]	CVPR'25	48.25	-	41.26	50.25	52.49	50.79	44.27	37.12	19.60
PL2-Transformer [57]	MM Asia'25	38.79	-	-	-	-	-	-	-	-
Point-CSA [58]	CVPR'25	38.87	-	33.47	41.45	43.55	42.35	35.02	29.43	13.62
PointNet4D [35]	WACV'26	51.24	-	44.74	54.47	56.08	55.24	46.34	33.89	13.29
PePNet	Ours	**54.63**	**61.14**	**50.73**	**57.42**	**62.75**	**58.75**	**49.37**	**41.28**	**22.15**

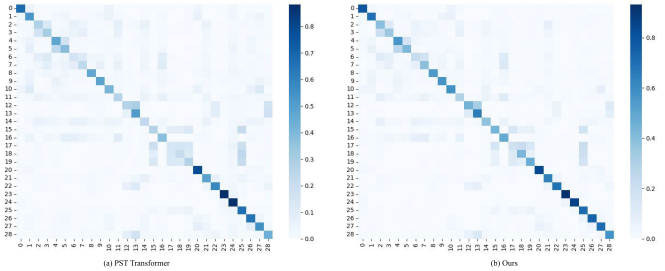


Fig. 8. Confusion matrix on the Momo test set. Left: PST Transformer. Right: Our method. Our method exhibits clearer class separation.

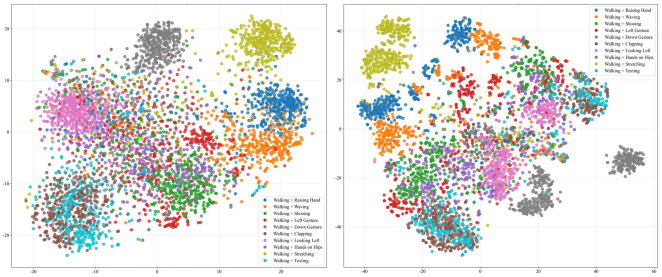


Fig. 9. t-SNE visualization of feature embeddings on the Momo test set. Left: PST Transformer. Right: Our method.

by + 3.39%. Compared to Mamba4D [32] (48.25%), 3DInAction [56] (46.23%), and PST-Transformer [55] (45.64%), PePNet yields absolute gains of 6.38%, 8.40%, and 8.99%, respectively. Other recent baselines, including Point-CSA [58] (38.87%), PL2-Transformer [57] (38.79%), and the skeleton-based A^2MC [53] (34.94%), fall further behind, confirming that neither Transformer pre-training nor skeleton-only representations can adequately address distance-induced sparsity without explicit pose guidance.

The distance-stratified breakdown further highlights PePNet's advantage. While PointNet4D performs competitively at near and mid ranges (56.08% at 10–15 m), it drops sharply beyond 25 m (33.89%) and 30 m (13.29%). PePNet maintains 41.28% and 22.15% at the same ranges, leading by + 7.39% and + 8.86%, demonstrating that pose-guided densification provides a complementary benefit that architectural advances alone cannot achieve. These results validate the effectiveness of integrating pose priors with Mamba-based spatio-temporal modeling for robust action recognition in long-range motion-in-motion scenarios.

2) Qualitative Analysis: To gain deeper insights into the performance improvements, we perform qualitative analyses using confusion matrices and feature visualizations.

a) Confusion matrix analysis: Fig. 8 compares the confusion matrix of our method with that of PST Transformer. We observe that PST Transformer exhibits noticeable confusion between spatially and motionally similar actions, such as Class 17 ('Thumbs Down'), Class 18 ('Shake Head'), and Class 19 ('Look Left'). Our method demonstrates significantly reduced confusion among these challenging classes. The diagonal elements corresponding to these classes are stronger, and off-diagonal misclassifications are diminished. This improvement highlights the benefit of the Human Pose Prior Enhancement module, which introduces pose-guided dynamic points, enhancing the model's ability to discriminate subtle differences in compound actions involving small upper-body movements combined with walking.

b) Feature visualization: We employ t-SNE [59] to visualize the feature embeddings learned by our method and PST Transformer on the test set, shown in Fig. 9. The visualization reveals that our method produces feature representations where different action classes form more distinct and compact clusters compared to PST-Transformer. While PST Transformer shows considerable overlap, particularly among classes involving upper-body movements during walking (e.g., 'Clap', 'Walk + Texting', 'Look Down', 'Look Left'), our method achieves better inter-class separability and intra-class cohesion. This visually confirms the enhanced feature representation capability of our approach, leading to improved discrimination and generalization for point cloud action recognition in complex environments.

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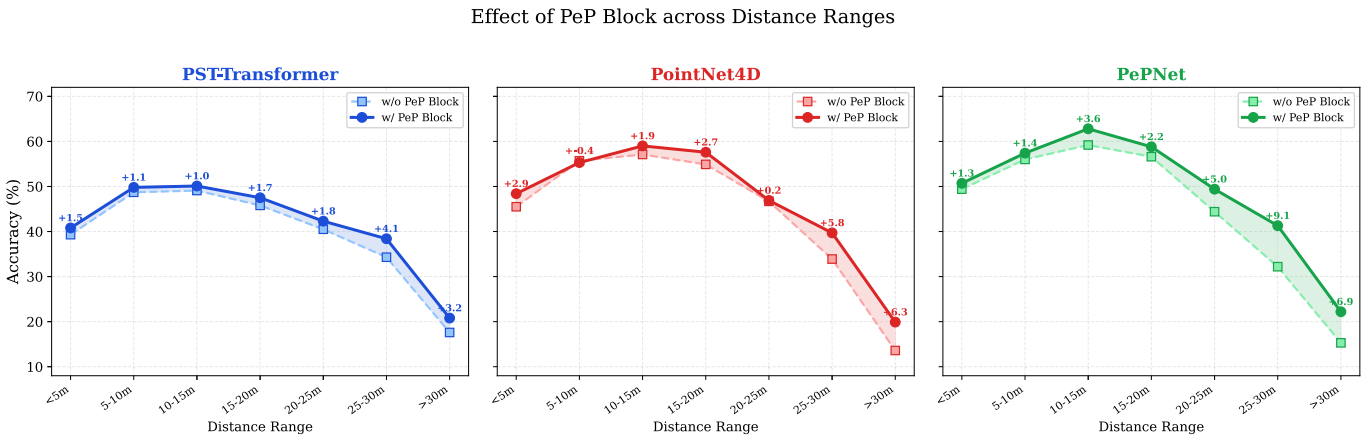


Fig. 10. Effect of the PeP Block across distance ranges on the Momo dataset, evaluated on three representative backbones: PST-Transformer (left), PointNet4D (middle), and PePNet (right). Solid lines denote models with the PeP Block integrated (*w/ PeP Block*), while dashed lines represent the corresponding baselines (*w/o PeP Block*). Annotated values indicate the absolute accuracy improvement (%) at each distance range.

C. Ablation Studies

We conduct ablation studies to dissect the contribution of individual components within our proposed framework.

1) Distance-Stratified Analysis of the PeP Block: To directly assess whether the PeP Block provides targeted benefits for long-range scenarios, we integrate it into three representative backbones — PST-Transformer, PointNet4D, and our full PePNet — and evaluate recognition accuracy across seven distance ranges on the Momo dataset. As shown in Fig. 10, the performance gain introduced by the PeP Block exhibits a clear distance-dependent trend across all three backbones: improvements are modest at short ranges (< 10 m) but grow substantially as distance increases. For PST-Transformer, the gain rises from + 1.5% at < 5 m to + 4.1% at 25–30 m and + 3.2% beyond 30 m. The effect is even more pronounced for PointNet4D, where the PeP Block yields + 6.3% at > 30 m compared to only + 2.9% at < 5 m. PePNet follows the same pattern, with gains scaling from + 1.3% at short range to + 9.1% at 25–30 m and + 6.9% beyond 30 m. This consistent distance-dependent behavior across architectures provides direct evidence that the PeP Block is not a generic enhancement module but is specifically effective under distance-induced point cloud sparsity: as subjects recede and point density degrades, the pose-guided structural augmentation and semantic edge weighting become increasingly critical for preserving the geometric cues necessary for reliable action recognition.

2) Generalization of the PeP Block Across Representation Paradigms: To address the concern of generalization across different point cloud representation methods, Table IV evaluates the effect of integrating the PeP Block into a diverse set of backbone architectures spanning four representative representation paradigms: convolution-based (PSTNet, PSTNet++), Transformer-based (PST-Transformer, Point-CSA), Mamba-Transformer hybrid (PointNet4D), and our Mamba-based PePNet. The results demonstrate consistent improvements across all backbones, confirming that the PeP Block functions as an architecture-agnostic enhancement module rather than one tailored to a specific representation strategy. The gains

TABLE IV

ACCURACY COMPARISON BEFORE AND AFTER ADDING THE HUMAN POSE PRIOR ENHANCEMENT MODULE TO VARIOUS BACKBONES

Method	Paradigm	w/o PeP (%)	w/ PeP (%)
PointNet++ [22]	MLP	23.86	25.95 (↑2.09)
PSTNet [9]	Conv	44.36	45.79 (↑1.43)
PSTNet++ [54]	Conv	45.58	46.88 (↑1.30)
PST-Transformer [55]	Transformer	45.64	47.86 (↑2.22)
3DInAction [56]	MLP	46.23	47.34 (↑1.11)
Point-CSA [58]	Transformer	38.87	40.05 (↑1.18)
PointNet4D [35]	Mamba-Trans.	51.24	53.50 (↑2.26)
PePNet (Ours)	**Mamba**	**51.36**	**54.63** (↑3.27)

range from + 1.18% for Point-CSA to + 2.26% for PointNet4D, with the largest relative improvements observed on simpler architectures such as PointNet++ (+ 2.09%), suggesting that the pose-guided structural augmentation is especially effective when the backbone's intrinsic feature extraction capacity is limited. Across Transformer-based backbones, PST-Transformer benefits by + 2.22%, while the Mamba-Transformer hybrid PointNet4D gains + 2.26%, indicating that the PeP Block provides complementary structural information that attention- and state-space-based feature extractors do not capture on their own. These results collectively demonstrate that the proposed pose-informed enhancement strategy is broadly applicable and consistently beneficial regardless of the underlying representation paradigm.

3) Analysis of the Dynamic Enhancement Mechanism: To provide deeper insight into the PeP Block, we visualize the learned distribution of the Dynamic Enhancement parameters before and after training. As shown in Fig. 11, while the fixed baseline adopts a uniform linear interpolation along skeletal edges, the trained model converges to a distinctly non-uniform sampling strategy. The difference heatmap reveals edge-specific patterns, where the network selectively allocates higher point density to certain skeletal segments — effectively functioning as a learned structural attention mechanism over the human body. This confirms that the Dynamic Enhancement Module operates as an intelligent, adaptive operator rather than a static upsampling tool: by reconfiguring point distributions

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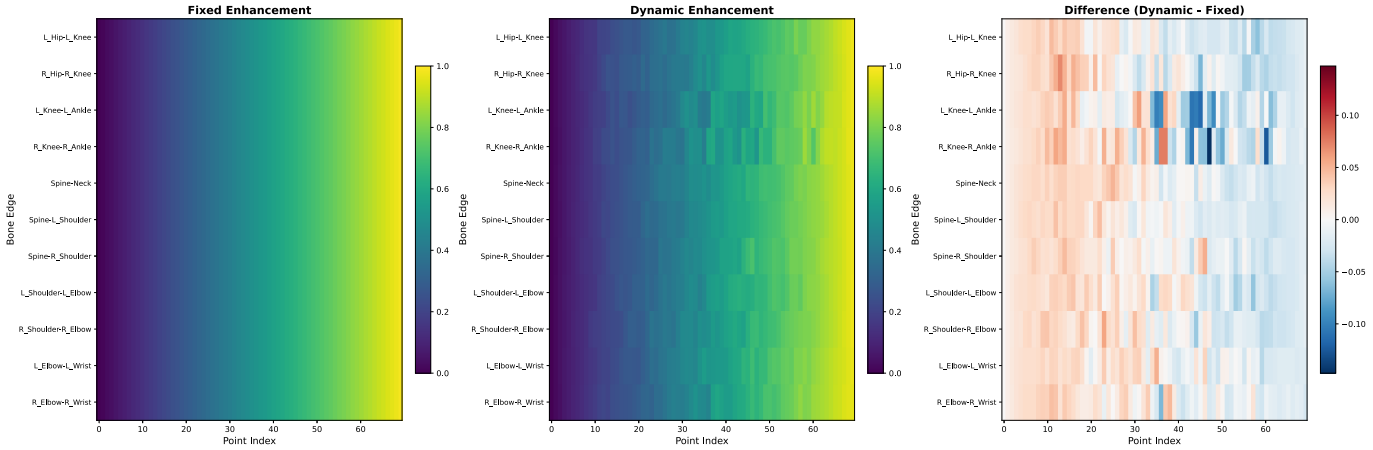


Fig. 11. Visualization of the Dynamic Enhancement Module weights distribution within the PeP Block. **Left:** Fixed Enhancement. **Middle:** The Dynamic Enhancement distribution after training. **Right:** Difference between Fixed and Dynamic Enhancement strategies. The varying patterns across different bone edges (y-axis) demonstrate that the PeP block dynamically optimizes point sampling density based on skeletal anatomy and action semantics.

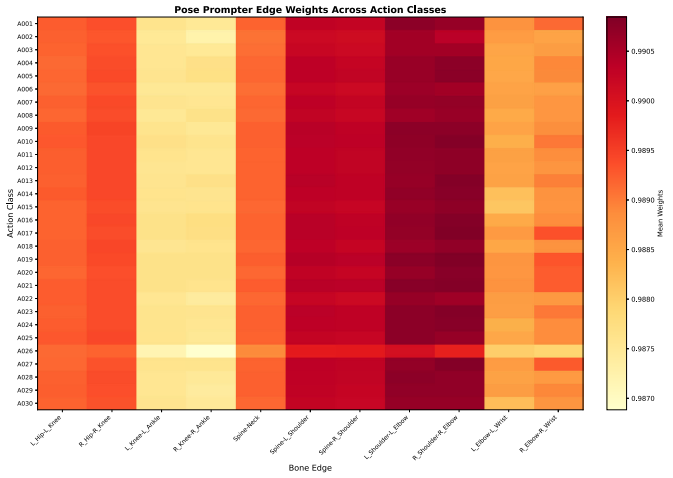


Fig. 12. Visualization of the learned Pose Prompter weights. The heatmap displays edge-wise importance across 30 action classes.

according to the current pose and bone scaling, it ensures that geometrically informative regions receive adequate representation even under severe distance-induced sparsity, directly contributing to the performance gains observed in long-range scenarios.

4) Analysis of Pose Prompter Mechanism: To investigate how the Pose Prompter guides body-region selection across different actions, we visualize the learned edge importance weights w_{edge} for all 30 action classes in Fig. 13. The heatmap reveals two consistent patterns. First, upper-limb segments (e.g., *Shoulder-Elbow*, *Spine-Shoulder*) receive substantially higher weights than lower-limb segments across most action classes, indicating that the network successfully suppresses the dominant walking motion — which carries little discriminative information — in favor of subtle upper-body articulations. Second, fine-grained weight variations are observable across classes: actions involving complex upper-body manipulations (e.g., *A028: Drinking*, *A029: Phone call*) exhibit heightened sensitivity to elbow and wrist-related edges. Together, these patterns confirm that the Pose Prompter functions as a semantic filter that selectively concentrates supplementary points on the most action-discriminative body regions, directly compensating for the loss of geometric detail caused by distance-induced sparsity.

5) Analysis of Adaptive Point Selection: To evaluate the behavior of the Adaptive Point Selection (PPS) module, we analyze the distribution of selection weights across three body regions — Upper Body, Lower Body, and Torso — for all 30 action classes, as shown in Fig. 13. The results reveal a consistent preference for upper-body segments (red bars) across all action categories, with selection weights substantially exceeding those of the lower body (green) and torso (blue). This pattern reflects two complementary properties of the learned selection strategy. On the one hand, the module concentrates supplementary points on upper-limb joints that carry the most discriminative semantic information (e.g., clapping, pointing, gesturing), effectively acting as a data-driven structural attention mechanism. On the other hand, it implicitly suppresses redundant contributions from the lower limbs despite their dominant walking motion, demonstrating that the differentiable sampling process learns to filter uninformative regions without explicit supervision. Together, these behaviors confirm that the PPS module not only reduces point redundancy but also preserves geometric fidelity in the most action-relevant body regions, contributing to robust recognition under the sparsity constraints of long-range outdoor LiDAR scenarios.

6) Comparison With Alternative Upsampling Strategies: To verify that the PeP Block's gains stem from pose-guided densification rather than increased point count alone, we compare it against four alternatives on the Momo dataset: No Upsampling, Random Upsampling, Midpoint Interpolation, and PUFM [60] (learning-based upsampling). Results are shown in Table V. Methods lacking skeletal guidance (Random, Midpoint) fall 1.78% and 2.60% below the PeP Block, as they densify irrelevant regions under long-range sparsity. PUFM achieves only 43.43%, barely above the no-upsampling baseline, due to distribution mismatch with sparse long-range

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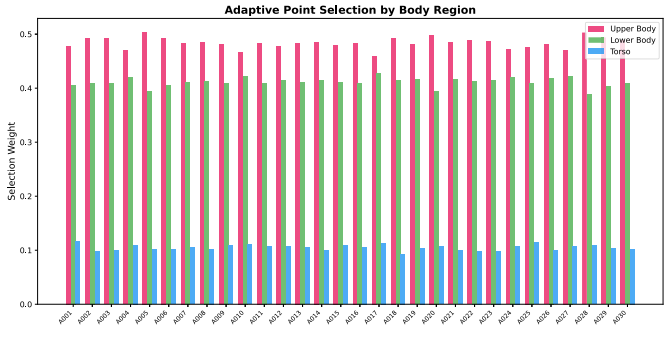


Fig. 13. Analysis of the Adaptive Point Selection (PPS) module. The bar chart illustrates the average selection weights assigned to different body regions (Upper Body, Lower Body, and Torso) across 30 action classes.

TABLE V
COMPARISON OF THE PEP BLOCK AGAINST ALTERNATIVE UPSAMPLING STRATEGIES ON THE MOMO DATASET

Method	Acc (%)	Infer Time (ms)
No Upsampling	42.36	0.14
Random Upsampling	51.71	0.28
Interpolation	50.89	0.65
PUFM [60]	43.43	52.18
PeP Block (Ours)	**54.63**	1.30

TABLE VI
IMPACT OF THE RATIO BETWEEN ORIGINAL AND ENHANCED POINTS WITHIN THE PEP BLOCK

PeP Block	Original Nums	Enhanced Nums	Accuracy (%)
×	1024	0	51.36
✓	0	1024	53.95
✓	256	768	54.20
✓	768	256	53.53
✓	**512**	**512**	**54.63**

inputs, while introducing 40× more latency (52.18 ms vs. 1.30 ms). The PeP Block achieves the best accuracy with negligible overhead, confirming that its advantage arises from semantically targeted, pose-guided densification.

7) Ratio of Original Vs. Enhanced Points: To determine the optimal fusion ratio between original points and pose-enhanced points, we performed experiments varying their respective counts while keeping the total number of points constant at 1024. The results are shown in Table VI. We observe three key findings: (1) The highest accuracy (54.63%) is achieved with a 1:1 ratio (512 original + 512 enhanced points). (2) Using only enhanced points (53.95%) outperforms using only original points (53.36%), confirming the strong representational power of the pose-guided generated points. (3) Fusing both types of points yields better performance than using either type alone, indicating that they provide complementary information. Original points retain direct geometric details, while enhanced points offer structured anatomical priors and potentially better coverage of key moving parts, leading to a more robust overall representation.

TABLE VII
IMPACT OF DIFFERENT COMPONENTS WITHIN THE PEP BLOCK

Dynamic Enhance	Pose Prompt	Adaptive Selection	Acc(%)
✓	×	×	52.36
✓	×	✓	53.80
×	✓	✓	53.69
✓	✓	✓	**54.63**

TABLE VIII
ACCURACY COMPARISON FOR DIFFERENT POINT SORTING STRATEGIES WITHIN THE SSM MODULE, USING SEQUENTIAL AND CROSS-TEMPORAL SCAN

Sequential Scan		Cross-Temporal Scan	
Sort Order	**Accuracy (%)**	**Sort Order**	**Accuracy (%)**
Unordered	49.96	Unordered	49.03
X	52.36	X	51.89
Y	52.59	Y	52.12
Z	54.63	Z	54.28
XYZ	54.88	XYZ	54.46
XZY	54.76	XZY	54.33
YXZ	54.21	YXZ	54.12
YZX	54.05	YZX	53.72
ZYX	55.38	ZYX	55.09
ZXY	**55.49**	**ZXY**	**55.18**

8) Synergy Within the Enhancement Module: We conducted a detailed ablation study, presented in Table VII, to understand the interplay between the core components of the PePNet: pose prompting, point cloud prompting (using point features to guide selection), and the adaptive selection mechanism (PPS). Applying only the pose-prompted dynamic enhancement yields 52.36% accuracy. Adding the adaptive selection mechanism (PPS) boosts performance to 53.80%. Combining both pose and point cloud prompts without PPS reaches 53.69%. The best performance (54.63%) is achieved when all components work synergistically. This demonstrates that the adaptive selection mechanism effectively refines the enhancement based on both point cloud features and pose priors. Furthermore, the combination of pose and point cloud prompts provides complementary information, leading to a performance gain (+ 2.27% compared to pose prompt only) greater than simply using one type of prompt, validating our multi-source information fusion strategy.

9) Impact of Human Body Longitudinal Scanning Strategies: Table VIII investigates the influence of different scanning strategies on our Mamba-based model performance. The results validate that our human body longitudinal scanning approach significantly outperforms unordered point processing (baselines around 49-50%). Our Z-axis prioritized scanning demonstrates clear effectiveness. Single-axis Z-ordering achieves 54.63% accuracy, confirming that aligning scanning with the natural head-to-toe structure of human poses better captures hierarchical motion patterns inherent in human actions. The bidirectional temporal scanning strategies further enhance performance. Both sequential scanning (maintaining longitudinal-temporal order) and cross-temporal scanning (establishing long-range dependencies) show comparable

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TABLE IX
COMPARISON OF LIDAR-BASED POSE ESTIMATION METHODS AS THE BACKBONE OF THE PEP BLOCK IN PEPNET

Pose Estimator	Source	Joints	Momo Acc (%)
SPiKE [42]	PR'24	15	53.12
LiDAR-HMR [41]	TMM'25	15	54.63

TABLE X
ROBUSTNESS OF PEPNET TO POSE PERTURBATIONS ON THE MOMO DATASET

Pose Perturb.	Overall	20–25m	25–30m	>30m
Clean pose	54.63	49.37	41.28	22.15
Gaussian 1%	53.11	46.01	32.78	11.96
Gaussian 3%	52.34	45.63	32.33	10.30
Gaussian 5%	45.45	39.20	22.52	9.30
Joint dropout 10%	52.64	45.40	33.00	13.29
Joint dropout 20%	52.44	46.20	31.22	11.63

effectiveness, with cross-temporal achieving 55.18% versus 55.49% for sequential scanning in the optimal ZXY configuration. Multi-axis combinations validate our longitudinal-first design. The ZXY strategy achieves optimal performance (55.49% sequential, 55.18% cross-temporal), followed closely by ZYX (55.38%, 55.09%). These results confirm that prioritizing the vertical axis while incorporating horizontal structure creates representations aligned with human anatomical organization. The substantial 5-6% improvement over unordered processing demonstrates that our longitudinal scanning enables the SSM to effectively capture the natural flow of human motion patterns, providing semantically meaningful sequence ordering for enhanced action recognition.

10) Comparison of LiDAR-Based Pose Estimation Backbones in PeP Block: Our PeP Block relies on a LiDAR-based human mesh recovery module to extract pose priors from raw point clouds. To investigate the sensitivity of our framework to the choice of pose estimator, we evaluate two representative LiDAR-based pose estimation methods, namely LiDAR-HMR [41] and SPiKE [42], as alternative backbones and report the results in Table IX. Under the same joint configuration (15 joints), LiDAR-HMR achieves 54.63% while SPiKE yields 53.12%, indicating that the accuracy and robustness of the pose estimator under long-range sparsity directly influence the quality of the geometric priors fed into the PeP Block. These results confirm two key findings. First, the quality of LiDAR-derived pose priors has a direct and measurable impact on final HAR performance, which validates the motivation for the pose-guided design of the PeP Block. Second, the framework is modular with respect to the pose estimation backbone and can seamlessly benefit from future advances in LiDAR-based human pose estimation.

11) Robustness to Pose Noise: We evaluate PePNet by perturbing external pose priors with Gaussian joint noise and random joint dropout. As shown in Table X, moderate perturbations are well tolerated: Gaussian noise up to 3% body scale causes at most a 2.29% drop, and 10%/20% joint dropout leads to 1.99%/2.19% decreases. Under severe 5% Gaussian noise, the overall accuracy drops by 9.18%, indicating that heavily corrupted pose priors can weaken pose-guided enhancement under extremely sparse observations.

TABLE XI
COMPUTATIONAL COST ANALYSIS OF EACH PROPOSED MODULE

Configuration	Params (M)	GFLOPs	Inference (ms)
Pose Estimation	1.74	5.09	27.47
PeP Block	0.31	6.94	0.74
Dynamic Enhancement	0.001	–	0.16
Adaptive Point Selection	0.306	6.94	0.57
ST-Tube Embedding	0.47	6.59	20.72
Mamba Block	20.62	0.34	4.59
Classification Head	1.11	0.01	0.06
PePNet (Full Model)	22.51	14.87	26.11

12) Computational Cost Analysis: Table XI reports the parameter count, FLOPs, and per-module inference time of each proposed component. The full PePNet comprises 22.51M parameters with a total inference time of 26.11 ms. The Pose Estimation module introduces 1.74M parameters and 27.47 ms latency as a one-time preprocessing step. Within the PeP Block (0.31M, 0.74 ms), the Adaptive Point Selection accounts for the majority of parameters (0.306M) and computation (6.94 GFLOPs), while the Dynamic Enhancement is nearly cost-free (0.001M, 0.16 ms). The ST-Tube Embedding dominates the runtime (20.72 ms) due to dynamic neighbor search over raw point clouds. Notably, the PeP Block introduces less than 3% of the total overhead while consistently improving recognition accuracy across multiple backbone architectures (Table IV), confirming a favorable trade-off between performance and computational cost.

VI. CONCLUSION

We tackle the challenge of distance variation in LiDAR-based Human Action Recognition. We introduce Momo, a large-scale dataset with actions captured at distances between 2 and 30 meters, under a "motion-in-motion" setting where both subjects and sensors are moving. This setup enables studying the effects of distance on point cloud quality and recognition accuracy. To reduce the influence of point cloud sparsity at longer distances, we propose PePNet, which enhances spatial representation using pose-guided features through a Dynamic Enhancement Module, a Pose Prompter, and an Adaptive Sampling Mechanism. Together with ST-Tube for local motion encoding and Mamba for sequence modeling, our method captures motion cues more effectively across varying distances. Experiments on the Momo dataset show clear performance gains over existing methods, particularly in distant scenarios where input data is sparse.

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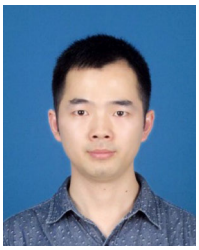
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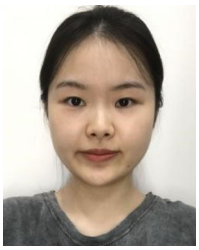
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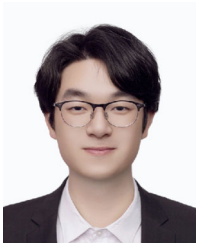
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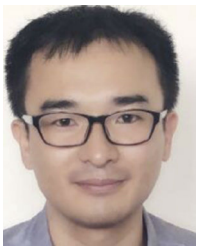
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